

COARSE GEOMETRY OF EXTENDED ADMISSIBLE GROUPS

TOAN TRONG DAO AND HOANG THANH NGUYEN

ABSTRACT. Extended admissible groups belong to a particular class of graphs of groups that admit a decomposition generalizing those of non-geometric 3-manifold groups and Croke-Kleiner admissible groups. In this paper, we study several coarse-geometric aspects of extended admissible groups. We show that changing the gluing edge isomorphisms does not affect the quasi-isometry type of these groups. We also prove that, under mild conditions on the vertex groups, extended admissible groups exhibit large-scale nonpositive curvature, thereby answering a question posed in [NY23].

As an application, our results enlarge the class of extended admissible groups known to admit well-defined quasi-redirecting boundaries, a notion recently introduced by Qing-Rafi. In addition, we compute the divergence of extended admissible groups, generalizing a result of Gersten from non-geometric 3-manifold groups to this broader setting. Finally, we study several aspects of subgroup structure in extended admissible groups.

1. INTRODUCTION

Let M be a non-geometric 3-manifold. The torus decomposition of M yields a nonempty minimal union $\mathcal{T} \subset M$ of disjoint essential tori, unique up to isotopy, such that each component M_v of $M \setminus \mathcal{T}$, called a *piece*, is either Seifert fibered or hyperbolic. There is an induced graph of groups decomposition $\mathcal{G}$ of $\pi_1(M)$ with underlying graph Γ as follows. For each piece M_v , there is a vertex v of Γ with vertex group $\pi_1(M_v)$. For each torus $T_e \in \mathcal{T}$ contained in the closure of pieces M_v and M_w , there is an edge e of Γ between vertices v and w . The associated edge group is $\pi_1(T_e) \cong \mathbb{Z}^2$ and the edge monomorphisms are the maps induced by inclusion. Each Seifert fibered piece M_v in the JSJ decomposition of M admits a Seifert fibration over a hyperbolic 2-orbifold Σ_v ; thus there is a short exact sequence

$$1 \rightarrow \mathbb{Z} \rightarrow \pi_1(M_v) \rightarrow \pi_1(\Sigma_v) \rightarrow 1$$

where $\mathbb{Z}$ is the normal cyclic subgroup of $\pi_1(M)$ generated by a fiber. If M_v is a hyperbolic piece, then $\pi_1(M_v)$ is hyperbolic relative to $\{\pi_1(T_1), \dots, \pi_1(T_\ell)\}$, where $\{T_1, \dots, T_\ell\}$ is the collection of boundary tori of M_v .

Motivated by this structure, Croke and Kleiner introduced the class of *admissible groups* in [CK02], abstracting the graph of groups structure of graph manifolds. In [MN24], the authors further introduced the class of *extended admissible groups*, which generalizes fundamental groups of all non-geometric 3-manifolds as well as Croke-Kleiner admissible groups. In an extended admissible group, vertex groups are allowed to be either central extensions of hyperbolic groups by $\mathbb{Z}$ or toral relatively hyperbolic groups, yielding a significantly broader and more flexible class of groups. For the precise definition of extended admissible groups, we refer the reader to Definition 2.7.

The large-scale geometry of extended admissible groups has recently attracted attention. The main result of [MN24] establishes quasi-isometric rigidity for this class, extending the seminal work of Kapovich-Leeb [KL97] on graph manifold groups. In [Ngu25], subgroup separability questions for extended admissible groups are studied. Property (QT) has been examined in [NY23], [HNY23], while in [HRSS22], the authors have shown that admissible groups are hierarchically hyperbolic

Date: January 1, 2026.

2010 *Mathematics Subject Classification.* 20F65, 20F67.

groups. In [ANR24], the authors demonstrate that admissible groups are $\mathcal{H}$ -inaccessible. Additionally, quasi-isometric rigidity is studied in [MN24], sublinearly Morse boundaries are studied in [NQ24], and quasi-redirecting boundaries are studied in [NQ25].

The present paper continues this line of research by examining several coarse-geometric and subgroup-theoretic properties of extended admissible groups, with the goal of extending classical results from non-geometric 3-manifold groups to this more general setting.

A natural problem accompanying quasi-isometric rigidity is quasi-isometric classification.

Question 1.1 (Quasi-isometric classification). Given a class $\mathcal{C}$ of finitely generated groups, determine when two elements of $\mathcal{C}$ are quasi-isometric.

Each vertex group of an extended admissible group is either a central extension of a hyperbolic group or is relatively hyperbolic; we call these type $\mathcal{S}$ and type $\mathcal{H}$ respectively. An extended admissible group G is called an *admissible group* if it has no vertex group of type $\mathcal{H}$. In [MN24], the authors show that quasi-isometries between extended admissible groups preserve vertex-group types and the quasi-isometry classes of the associated hyperbolic quotients. This result implies that there are infinitely many quasi-isometry classes of admissible and extended admissible groups. However, it leaves open an important structural question: to what extent does the choice of edge gluing isomorphisms influence the quasi-isometry type? This motivates the following question.

Question 1.2. To what extent do the gluing edge isomorphisms influence the quasi-isometry type of the resulting admissible groups?

Our first result in this paper gives a positive answer to Question 1.2.

Theorem 1.3. *Let $\mathcal{G}$ and $\mathcal{G}'$ be admissible graphs of groups with identical underlying graph, vertex groups, and edge groups, differing only in their edge isomorphisms. Then their fundamental groups are quasi-isometric.*

This result shows that, within the class of admissible groups, the large-scale geometry is insensitive to the specific gluing data, paralleling classical results for graph manifold groups.

Motivated by the notion “flip graph manifolds” introduced by Kapovich–Leeb [KL98], we next study admissible groups acting geometrically on Hadamard spaces via so-called CKA actions. In particular, we consider *flip CKA actions*, which arise from specific choices of edge identifications (see Definition 4.2). These actions play a central role in understanding large-scale nonpositive curvature phenomena.

A key ingredient in this analysis is the notion of omnipotence (see Definition 4.3), introduced by Wise [Wis00] which has been widely used in subgroup separability. Many familiar classes of groups, including free groups, surface groups, Fuchsian groups, and virtually special hyperbolic groups, are omnipotent. Using omnipotence assumptions on the hyperbolic quotients of vertex groups, we answer a question posed in [NY23].

Theorem 1.4. *Let G be an admissible group such that each vertex group is a central extension of an omnipotent hyperbolic $\text{CAT}(0)$ group by $\mathbb{Z}$. Then G is quasi-isometric to a group admitting a flip CKA action.*

As a consequence, we obtain new information about large-scale curvature invariants of extended admissible groups. In particular, we compute their divergence. Divergence is a quasi-isometry invariant introduced by Gersten [Ger94], which plays a key role in distinguishing geometric behaviors of groups. Gersten showed that non-geometric 3-manifold groups have either quadratic or exponential divergence, depending on the presence of hyperbolic pieces. We extend this result to extended admissible groups under mild hypotheses on vertex groups.

Corollary 1.5. *Let G be an extended admissible group such that for each vertex group G_v of type $\mathcal{S}$, its non-elementary hyperbolic factor Q_v is omnipotent and is a $CAT(0)$ group. Then the divergence of G is quadratic if and only if G contains no vertex groups of type $\mathcal{H}$ and it is exponential otherwise.*

In particular, assume that G' is another extended admissible group satisfying the same conditions as G . If G contains no vertex groups of type $\mathcal{H}$ and G' contains at least one vertex group of type $\mathcal{H}$, then G and G' are not quasi-isometric.

Another application concerns quasi-redirecting boundaries, recently introduced by Qing and Rafi [QR24] as a candidate for a quasi-isometry invariant boundary theory extending the Gromov boundary. While the existence of such boundaries is known for several important classes of groups, it remains open in general. Our main theorem allows the extension of the construction in [NQ25] from groups admitting flip CKA actions to a broader class of admissible and extended admissible groups.

Corollary 1.6. *Let G be an extended admissible group such that for each vertex group G_v of type $\mathcal{S}$, its non-elementary hyperbolic factor Q_v is omnipotent and is a $CAT(0)$ group. Then G has well-defined quasi-redirecting boundary.*

In particular, we obtain new examples among free-by-cyclic groups, a class that has been extensively studied but for which the existence of well-defined quasi-redirecting boundaries was previously unknown.

Corollary 1.7. *Let Φ be a linearly growing automorphism of the finite rank free group F and let $G = F \rtimes_{\Phi} \langle t \rangle$ be its mapping torus. Suppose that G is unbranched in the sense of [BGH25]. Then G has well-defined quasi-redirecting boundary.*

Finally, we investigate subgroup structure in extended admissible groups. We study the relationship between strong quasiconvexity (see Definition 5.6), finite height, and (virtual) malnormality (see Definition 5.5). Building on work of Tran [Tra19] and Hruska–Wise [HW09], we show that separable, strongly quasiconvex subgroups are virtually almost malnormal in finitely generated groups. As an application, we characterize strongly quasiconvex subgroups of graph manifold groups as precisely those that are virtually malnormal.

Theorem 1.8. *Let H be a separable, strongly quasiconvex subgroup of a finitely generated group G . Then there is a finite index subgroup K of G containing H such that H is almost malnormal in K . Furthermore, suppose that G is virtually torsion-free then H is virtually malnormal.*

Suppose G is the fundamental group of a graph 3-manifold M . Then a finitely generated subgroup H of $\pi_1(M)$ is strongly quasiconvex if and only if H is virtually malnormal in G .

We conclude by establishing several embedding obstructions for extended admissible groups, including consequences of the Rapid Decay property and examples of non-subgroup-separable behavior.

Proposition 1.9. *Let G be an extended admissible group. Then*

- (1) *G has Rapid Decay property. As a consequence, amenable groups with exponential growth, Thompson’s groups, $SL_n(\mathbb{Z})$ with $n > 3$, intermediate growth groups, Baumslag-Solitar group $BS(p, q)$ (where $p \neq q$) cannot be embedded as subgroups of G .*
- (2) *Consider the following Croke–Kleiner group $L = \langle i, j, k, l \mid [i, j], [j, k], [k, l] \rangle$. Suppose that G contains at least one maximal admissible component then there is an embedding $L \rightarrow G$ and hence G is not subgroup separable.*

1.1. Overview. In Section 2 we recall the necessary background on trees of spaces and extended admissible groups. Section 3 proves the quasi-isometric invariance under changing edge maps. In Section 4 we establish large-scale $CAT(0)$ geometry via flip CKA actions and derive consequences for

divergence and boundaries. Section 5 studies subgroup structure, focusing on strong quasiconvexity and malnormality.

2. PRELIMINARIES

In this section, we review some concepts in geometric group theory that will be used throughout the paper.

2.1. Coarse geometry.

Definition 2.1. Let X and Y be metric spaces and f be a map from X to Y .

- (1) We say that f is a (K, A) -quasi-isometric embedding if for all $x, x' \in X$,

$$\frac{1}{K}d(x, x') - A \leq d(f(x), f(x')) \leq Kd(x, x') + A.$$

- (2) We say that f is a (K, A) -quasi-isometry if it is a (K, A) -quasi-isometric embedding such that $Y = N_A(f(X))$.

- (3) We say two quasi-isometries $f, g: X \rightarrow Y$ are A -close if

$$\sup_{x \in X} d_Y(f(x), g(x)) \leq A$$

and are close if they are A -close for some $A \geq 0$.

2.2. Tree of spaces of graph of groups. We assume familiarity with Bass–Serre theory; see [SW79] for details. However, to fix notation and terminology, we give some brief definitions.

We first establish some terminology regarding graphs. A *graph* Γ consists of a set $V\Gamma$ of vertices, a set $E\Gamma$ of oriented edges, and maps $\iota, \tau: E\Gamma \rightarrow V\Gamma$. There is a fixed-point free involution $E\Gamma \rightarrow E\Gamma$, taking an edge $e \in E\Gamma$ such that $\iota e = v$ and $\tau e = w$ to an edge $\bar{e}$ satisfying $\iota \bar{e} = w$ and $\tau \bar{e} = v$. We also write e_+ and e_- to denote τe and ιe respectively. An *unoriented edge* of Γ is the pair $\{e, \bar{e}\}$. If v is a vertex, we define $\text{Link}(v) = \{e \in E\Gamma \mid e_- = v\}$.

Each connected graph can be identified with a metric space by equipping its topological realization with the path metric in which each edge has length one. A *combinatorial path* in X is a path $p: [0, n] \rightarrow X$ for some $n \in \mathbb{N}$ such that for every integer i , $p(i)$ is a vertex, and $p|_{[i, i+1]}$ is either constant or traverses an edge of X at unit speed. Every geodesic between vertices of X is necessarily a combinatorial path.

Definition 2.2. A *graph of groups* $\mathcal{G} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ consists of the following data:

- (1) a graph Γ (called the *underlying graph*),
- (2) a group $G_{\hat{v}}$ for each vertex $\hat{v} \in V(\Gamma)$ (called a *vertex group*),
- (3) a subgroup $G_{\hat{e}} \leq G_{\hat{e}_-}$ for each edge $\hat{e} \in E(\Gamma)$ (called an *edge group*),
- (4) an isomorphism $\tau_{\hat{e}}: G_{\hat{e}} \rightarrow G_{\hat{e}_-}$ for each $\hat{e} \in E(\Gamma)$ such that $\tau_{\hat{e}}^{-1} = \tau_{\bar{\hat{e}}}$ (called an *edge map*).

The *fundamental group* $G = \pi_1(\mathcal{G})$ of a graph of groups $\mathcal{G}$ is as defined in [SW79]. Via the construction of G , we will always view vertex and edge groups of $\mathcal{G}$ as subgroups of G .

We use the following notation for trees of spaces, similar to [CM17].

Definition 2.3. A *tree of spaces* $X := X(T, \{X_v\}_{v \in V(T)}, \{X_e\}_{e \in E(T)}, \{\alpha_e\}_{e \in E(T)})$ consists of:

- (1) a simplicial tree T , called the *base tree*;
- (2) a connected graph X_v for each vertex v of T , called a *vertex space*;
- (3) a connected subgraph $X_e \subseteq X_{e_-}$ for each oriented edge e (with the initial vertex denoted by e_-) of T , called an *edge space*;
- (4) graph isomorphisms $\alpha_e: X_e \rightarrow X_{\bar{e}}$ for each edge $e \in ET$, such that $\alpha_{\bar{e}} = \alpha_e^{-1}$.

Metrics on X : We think of X as a metric space by equipping it with the path metric. Each vertex and edge space X_x of X with $x \in VT \sqcup ET$ is thus endowed with two a priori different metrics: the induced path metric on X_x , and the subspace metric when X_x is considered as a subspace of X .

2.3. Tree of spaces from graph of groups. We now explain how to associate a tree of spaces to a graph of finitely generated groups.

Let $\mathcal{G} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ be a graph of finitely generated groups and let G be the fundamental group of this graph of groups. We recall the associated Bass–Serre tree T is constructed so that vertices (resp. edges) of T correspond to left cosets of vertex (resp. edge) groups of $\mathcal{G}$.

We now describe a tree of spaces X . For each $\hat{x} \in VT \sqcup ET$, we fix a finite generating set $J_{\hat{x}}$ of $G_{\hat{x}}$, chosen such that $\tau_{\hat{e}}(J_{\hat{e}}) = J_{\hat{e}_+}$, and $J_{\hat{e}} \subseteq J_{\hat{v}}$ if $\hat{e} \in ET$ with $\iota\hat{e} = \hat{v}$. We now define a graph W with vertex set $VT \times G$ and edge set

$$\{((\hat{v}, g), (\hat{v}, gs)) \mid g \in G, s \in J_{\hat{v}}\}.$$

The components of W are in bijective correspondence with left cosets of vertex groups of $\mathcal{G}$, and hence with vertices of T . If $v \in VT$ corresponds to $gG_{\hat{v}}$, we define X_v to be the component of W with vertex set $\{(\hat{v}, h) \mid h \in gG_{\hat{v}}\}$. We note that the component of W corresponding to a coset $gG_{\hat{v}}$ is isometric to the Cayley graph of $G_{\hat{v}}$ with respect to $J_{\hat{v}}$.

Suppose $e \in ET$ corresponds to a coset $gG_{\hat{e}}$. By the definition of T , if $\hat{v} = \hat{e}_-$ and $\hat{w} = \hat{e}_+$, then $v := e_-$ and $w := e_+$ correspond to the cosets $gG_{\hat{v}}$ and $gG_{\hat{w}}$. We define the edge space X_e to be the graph with vertex set

$$\{(\hat{v}, h) \mid h \in gG_{\hat{e}}\} \subseteq X_v$$

and edge set

$$\{((\hat{v}, h), (\hat{v}, hs)) \mid h \in gG_{\hat{e}}, s \in J_{\hat{e}}\}.$$

Thus X_e is isomorphic to the Cayley graph of $G_{\hat{e}}$ with respect to $J_{\hat{e}}$. The attaching map $\alpha_e : X_e \rightarrow X_v$ is defined by $\alpha_e : (\hat{v}, h) \mapsto (\hat{v}, g\tau_{\hat{e}}(g^{-1}h))$ on vertices, and similarly on edges, where $\tau_{\hat{e}} : G_{\hat{e}} \rightarrow G_{\hat{e}_+} \leq G_{\hat{w}}$ is the edge map of $\mathcal{G}$.

Definition 2.4. Given a graph of finitely generated groups $\mathcal{G}$, the tree of spaces X constructed above is *the tree of spaces associated with the graph of groups $\mathcal{G}$* .

The tree of spaces X is a proper geodesic metric space (see Lemma 2.13 of [CM17]). The natural action of G on W (fixing the VT factor) induces an action of G on X . Applying the Milnor–Schwarz lemma we deduce:

Proposition 2.5 (Section 2.5 of [CM17]). *Suppose G , T and X are as above. Then there exists a quasi-isometry $f : G \rightarrow X$ and $A \geq 0$ such that $d_{\text{Haus}}(f(gG_{\hat{x}}), X_x) \leq A$ for all $x \in VT \sqcup ET$, where x corresponds to the coset $gG_{\hat{x}}$.*

The following theorem explains how to build a quasi-isometry between trees of spaces by patching together quasi-isometries of vertex spaces. This can be done if quasi-isometries on adjacent vertex spaces agree up to a uniformly bounded error on their common edge space.

Theorem 2.6. [CM17, Corollary 2.16] *Let $K > 1$ and $A > 0$. Suppose that $X := X(T, \{X_v\}, \{X_e\}, \{\alpha_e\})$ and $X' := X'(T', \{X'_v\}, \{X'_e\}, \{\alpha'_e\})$ are trees of spaces, and that there is a tree isomorphism $\xi : T \rightarrow T'$. Suppose for every $v \in V(T)$ and $e \in E(T)$ there is a (K, A) -quasi-isometry $\phi_v : X_v \rightarrow X'_{\xi(v)}$ and $\phi_e : X_e \rightarrow X'_{\xi(e)}$. Suppose also that for every $e \in E(T)$, the following diagrams commute*

up to uniformly bounded error A .

$$\begin{array}{ccc} X_e & \xrightarrow{\phi_e} & X'_{\xi(e)} \\ \downarrow & & \downarrow \\ X_{e-} & \xrightarrow{\phi_{e-}} & X'_{\xi(e-)} \end{array} \quad \begin{array}{ccc} X_e & \xrightarrow{\phi_e} & X'_{\xi(e)} \\ \downarrow \alpha_e & & \downarrow \alpha'_{\xi(e)} \\ X_{e+} & \xrightarrow{\phi_{e+}} & X'_{\xi(e+)} \end{array}$$

Then there is a quasi-isometry $\phi : X \rightarrow X'$ such that $\phi|_{X_v} = \phi_v$ for every $v \in V(T)$.

2.4. Extended admissible groups. We now define the class of extended admissible groups.

Definition 2.7. A group G is an *extended admissible group* if it is the fundamental group of a graph of groups $\mathcal{G}$ such that:

- (1) The underlying graph Γ of $\mathcal{G}$ is a connected finite graph with at least one edge, and every edge group is $\mathbb{Z}^2$.
- (2) Each vertex group G_v is one of the following two types:
 - (a) Type $\mathcal{S}$: G_v has center $Z_v := Z(G_v) \cong \mathbb{Z}$ such that the quotient $Q_v := G_v/Z_v$ is a non-elementary hyperbolic group. We call Z_v and Q_v the *kernel* and *hyperbolic quotient* of G_v respectively.
 - (b) Type $\mathcal{H}$: G_v is hyperbolic relative to a collection $\mathbb{P}_v$ of virtually $\mathbb{Z}^2$ -subgroups, where all edge groups incident to G_v are contained in $\mathbb{P}_v$, and G_v doesn't split relative to $\mathbb{P}_v$ over a subgroup of an element of $\mathbb{P}_v$.
- (3) For each vertex group G_v , if $e, e' \in \text{Link}(v)$ and $g \in G_v$, then $gG_e g^{-1}$ is commensurable to $G_{e'}$ if and only if both $e = e'$ and $g \in G_e$.
- (4) For every edge group G_e such that G_{e-} and G_{e+} are vertex groups of type $\mathcal{S}$, the subgroup generated by $\tau_e(Z_{e+} \cap G_e)$ and $Z_{e-} \cap G_e$ has finite index in G_e .

Definition 2.8. An extended admissible group G is called an *admissible group* if it has no vertex group of type $\mathcal{H}$.

Convention: For the rest of this paper, if G is an extended admissible group, we will assume that all the data $\mathcal{G}, G_{\hat{v}}, Z_{\hat{v}}, Q_{\hat{v}}$, etc. in Definition 2.7 are fixed, and will make use of this notation without explanation. If G' is another admissible group, we use the notation $\mathcal{G}', G'_{\hat{v}}, Z'_{\hat{v}}, Q'_{\hat{v}}$ etc.

Below are some examples of extended admissible groups.

Example 2.9. (1) (3-manifold groups) The fundamental group of a compact, orientable, non-geometric, irreducible 3-manifold M with empty or toroidal boundary is an extended admissible group. Seifert fibered and hyperbolic pieces correspond to type $\mathcal{S}$ and $\mathcal{H}$ vertex respectively. Fundamental groups of graph manifolds are admissible groups.

- (2) (Torus complexes) Let $n \geq 3$ be an integer. Let $T_1, T_2, \dots, T_n$ be a family of flat two-dimensional tori. For each i , we choose a pair of simple closed geodesics a_i and b_i such that $a_i \cap b_i \neq \emptyset$ and $\text{length}(b_i) = \text{length}(a_{i+1})$, identifying b_i and a_{i+1} and denote the resulting space by X . For each $i \in \{1, \dots, n-1\}$, we denote $V_i := T_i \cup T_{i+1} / \{b_i = a_{i+1}\}$. Let $S_i^1 \subset V_i$ be the subspace of V_i obtained by gluing b_i to a_{i+1} . The space X is obtained by gluing each V_i to V_{i+1} via the gluing map

$$\tau_i : b_{i+1} \times S_i^1 \subset V_i \rightarrow a_{i+1} \times S_{i+1}^1 \subset V_{i+1}$$

by sending $b_{i+1} \rightarrow S_{i+1}^1$ and $S_i^1 \rightarrow a_{i+1}$ accordingly. Such a gluing map is called a “flip” map in the literature.

Note that V_i is homotopic equivalent to the product of S_i^1 with the wedge of two circles a_i and b_{i+1} . The fundamental group $G = \pi_1(X)$ has a graph of groups structure where each vertex group $\pi_1(V_i) = (\langle a_i \rangle * \langle b_{i+1} \rangle) \times \mathbb{Z} = F_2 \times \mathbb{Z}$, edge groups are $\mathbb{Z}^2$ and edge maps are

induced by the gluing maps τ_i . It is clear that with this graph of groups structure, $\pi_1(X)$ is an admissible group.

Note that our space X is a local CAT(0) space, and hence the universal cover $\tilde{X}$ is CAT(0) by the Cartan-Hadamard theorem. This space is studied in [CK00].

Lemma 2.10. [HRSS22, Lemma 4.2] *Let $\mathcal{G} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ be an admissible group. Each vertex group $G_{\hat{v}}$ has an infinite generating set $S_{\hat{v}}$ so that the following holds.*

- (1) *The Cayley graph $\text{Cay}(G_{\hat{v}}, S_{\hat{v}})$ is quasi-isometric to a line.*
- (2) *The inclusion map $Z_{\hat{v}} \rightarrow \text{Cay}(G_{\hat{v}}, S_{\hat{v}})$ is a $Z_{\hat{v}}$ -equivariant quasi-isometry.*

Remark 2.11. Without loss of generality, we can assume that the finite generating set $J_{\hat{v}}$ of $G_{\hat{v}}$ is contained in $S_{\hat{v}}$.

Recall from the construction in Section 2.2 that each vertex space X_v of X is identified with the Cayley graph of a vertex group $G_{\hat{v}}$ of $\mathcal{G}$ with respect to some generating set $J_{\hat{v}}$.

Definition 2.12. (Subspace L_v and $\mathcal{H}_v$) Suppose that $v \in T$ corresponds to a coset $gG_{\hat{v}}$. Let $L_v \subset X_v$ be the graph with vertex set $gG_{\hat{v}}$ and with an edge connecting $x, y \in gG_{\hat{v}}$ if $x^{-1}y \in S_{\hat{v}}$. In particular, L_v is isometric to $\text{Cay}(G_{\hat{v}}, S_{\hat{v}})$, which is a quasi-line by Lemma 2.10.

Let $\mathcal{H}_v$ be the graph with vertex set $gG_{\hat{v}}$ and an edge connecting $x, y \in gG_{\hat{v}}$ if $x^{-1}y \in J_{\hat{v}} \cup Z_{\hat{v}}$. It is isometric to $\text{Cay}(G_{\hat{v}}, J_{\hat{v}} \cup Z_{\hat{v}})$. We call $\mathcal{H}_v$ is the *quotient space* of X_v .

Remark 2.13. For any $g \in G$ and for each vertex $v \in V(T)$ we have $gL_v = L_{gv}$.

Definition 2.14 (Quotient maps, boundary lines). Suppose that $v \in T$ corresponds to a coset $gG_{\hat{v}}$. Since L_v and $\mathcal{H}_v$ are each obtained from X_v by adding extra edges, there are distance non-increasing maps $p_v: X_v \rightarrow L_v$ and $\pi_v: X_v \rightarrow \mathcal{H}_v$ that are the identity on vertices. We call such $\pi_v: X_v \rightarrow \mathcal{H}_v$ is a *quotient map*. For each $e \in E(T)$ with $v = e_-$, we define the *boundary line* ℓ_e of $\mathcal{H}_v$ associated to e is

$$\ell_e := \pi_v(X_e) \subseteq \mathcal{H}_v.$$

Let w be an adjacent vertex of v and denote the oriented edge $[v, w]$ by e . Let

$$\psi_e: \ell_{\bar{e}} \rightarrow L_v$$

be the restriction to the boundary line $\ell_{\bar{e}}$ of the composition $p_v \circ \alpha_{\bar{e}} \circ \pi_w^{-1}$.

Remark 2.15. (1) The space $\mathcal{H}_v$ is constructed to represent the geometry of $Q_{\hat{v}} = G_{\hat{v}}/Z_{\hat{v}}$ and is relatively hyperbolic to the collection $\{\ell_e\}_{e \rightarrow v}$ (see [HRSS22, Lemma 2.15]).
(2) It is proved in [ANR24, Lemma 2.18] that ψ_e is a uniform quasi-isometry. Namely, there exists constants $\lambda > 1, c \geq 0$ such that for each oriented edge e in T then $\psi_e: \ell_{\bar{e}} \rightarrow L_v$ is a (λ, c) -quasi-isometry.

Lemma 2.16. *There exist constants $\lambda \geq 1, c \geq 0$ such that the following holds. Suppose that $v \in T$ corresponds to a coset $gG_{\hat{v}}$ where $\hat{v}$ is a vertex in the underlying graph Γ . Consider the map*

$$f_v: X_v \rightarrow \mathcal{H}_v \times L_v$$

defined by $x \mapsto (\pi_v(x), p_v(x))$ where π_v and p_v are maps given by Definition 2.14. Then f_v is a (λ, c) -quasi-isometry.

Proof. We consider two natural actions $G_{\hat{v}} \curvearrowright Q_{\hat{v}}$ and $G_{\hat{v}} \curvearrowright L_{\hat{v}} := \text{Cay}(G_{\hat{v}}, S_{\hat{v}})$ of $G_{\hat{v}}$ on quotients $Q_{\hat{v}}$ and the quasi-line $\text{Cay}(G_{\hat{v}}, S_{\hat{v}})$ respectively. It is shown in [HRSS22, Corollary 4.3] that the diagonal action $G_{\hat{v}} \curvearrowright Q_{\hat{v}} \times L_{\hat{v}}$ is metrically proper and co-bounded, and hence the orbit map (with respect to a fixed basepoint) denoted by $f_{\hat{v}}: G_{\hat{v}} \rightarrow Q_{\hat{v}} \times L_{\hat{v}}$ is a quasi-isometry such that the composition of $f_{\hat{v}}$ with the projection $Q_{\hat{v}} \times L_{\hat{v}} \rightarrow Q_{\hat{v}}$ is the quotient map $q_{\hat{v}}: G_{\hat{v}} \rightarrow Q_{\hat{v}} = G_{\hat{v}}/Z_{\hat{v}}$. It implies that f_v is a quasi-isometry. Since there are finitely many vertices in the underlying graph Γ , we conclude that f_v is a quasi-isometry with uniform quasi-isometric constants λ, c . $\square$

3. CHANGING EDGE MAPS DOES NOT CHANGE QUASI-ISOMETRIC TYPE

In this section, we are going to prove Theorem 1.3 by showing that if two admissible groups $G = (\Gamma, \{G_{\hat{v}_i}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ and $G' = (\Gamma, \{G'_{\hat{v}_i}\}, \{G'_{\hat{e}}\}, \{\sigma_{\hat{e}}\})$ differ only in their edge isomorphisms then $G = \pi_1(\mathcal{G})$ and $G' = \pi_1(\mathcal{G}')$ are quasi-isometric. Fix trees of spaces (X, T) , (X', T) associated with admissible groups G and G' respectively, with the same associated Bass–Serre tree T . By Proposition 2.5, G and G' are quasi-isometric to X and X' respectively. Hence it suffices to show that X and X' are quasi-isometric. To do so, we are going to construct collections of quasi-isometries

$$\{\phi_v : X_v \rightarrow X'_v\}_{v \in V(T)} \quad \text{and} \quad \{\phi_e : X_e \rightarrow X'_e\}_{e \in E(T)}$$

between the vertex and edge spaces in tree of spaces so that these collections of maps satisfy conditions Theorem 2.6.

Outline of the proof: In Section 3.1, using a fixed choice of quasi-isometries on representatives of vertex orbits, we extend these maps equivariantly to all vertex spaces and, by restriction, obtain induced quasi-isometries on edge spaces. The main technical step is to verify that the resulting vertex and edge quasi-isometries satisfy the compatibility conditions of Theorem 2.6, despite the change in edge isomorphisms. This is done in Section 3.2.

3.1. Construction of vertex/edge maps. Since G and G' share the same Bass–Serre tree T , their vertex spaces X_v and X'_v (and similarly, the edge spaces X_e and X'_e) are naturally identified. The only difference is in the gluing isomorphisms (from τ_e to σ_e). Let $G_{\hat{v}_0}, \dots, G_{\hat{v}_m}$ be the vertex subgroups of G . We proceed as follows:

Choice of Transversals: For each i , we fix $\mathfrak{S}_{\hat{v}_i}$ a set of transversals for left cosets of $G_{\hat{v}_i}$ in G such that $1 \in \mathfrak{S}_{\hat{v}_i}$.

Base quasi-isometries: For each vertex v in $V(T)$, let L_v and L'_v be the spaces defined in Definition 2.12 with respect to tree of spaces X and X' . Given a vertex $\hat{v}_i \in \{\hat{v}_0, \hat{v}_1, \dots, \hat{v}_m\}$ in the underlying graph Γ , we fix a vertex $v_i \in V(T)$ corresponding to $G_{\hat{v}_i} = 1 \cdot G_{\hat{v}_i}$. Recall from Definition 2.12 that L_{v_i} (resp L'_{v_i}) is the graph with vertex set $1 \cdot G_{\hat{v}_i}$ with an edge connecting $x, y \in 1 \cdot G_{\hat{v}_i}$ if $x^{-1}y \in S_{\hat{v}_i}$. In particular, $L_{v_i} = L'_{v_i}$. We also recall two quotient spaces $\mathcal{H}_{v_i}$ and $\mathcal{H}'_{v_i}$ from Definition 2.12 as well and remark that $\mathcal{H}_{v_i} = \mathcal{H}'_{v_i}$. We fix a quasi-isometry $\zeta_{v_i} : L_{v_i} \rightarrow L'_{v_i}$ which is the composition:

$$L_{v_i} \rightarrow X_{v_i} \rightarrow \mathcal{H}_{v_i} \times L_{v_i} = \mathcal{H}'_{v_i} \times L'_{v_i} \rightarrow L'_{v_i}$$

where the first map is the inclusion of L_{v_i} to X_{v_i} (as $L_v \subset X_v$), the second map is f_{v_i} given by Lemma 2.16 and the third map is the natural projection of $\mathcal{H}'_{v_i} \times L'_{v_i}$ into its second factor.

Extending to all vertices: At the moment, we have defined maps $\zeta_{v_0}, \zeta_{v_1}, \dots, \zeta_{v_m}$. We need to define ζ_v for an arbitrary vertex v in $V(T)$. For each vertex $v \in V(T)$, there exists a vertex $\hat{v}_i \in \{\hat{v}_0, \hat{v}_1, \dots, \hat{v}_m\}$ and a group element $t \in \mathfrak{S}_{\hat{v}_i}$ such that v corresponds to the coset $tG_{\hat{v}_i}$.

We write t in reduced form relative to a fixed maximal tree $\Lambda \subset \Gamma$. Namely, fix a maximal tree $\Lambda \subset \Gamma$. G has a finite generating set of the form $\mathcal{S} = \cup_{i=1}^m J_{\hat{v}_i} \cup J_0$ where J_0 consists of stable letters t_e corresponding to edges outside the maximal tree Λ (and $t_e = 1$ when $e \in E(\Lambda)$). Similarly for G' with the same maximal tree Λ . We first write the group element $t \in \mathfrak{S}_{\hat{v}_i}$ in reduced form

$$t = g_0 t_{\alpha_1} g_1 t_{\alpha_2} \dots t_{\alpha_k} g_k$$

where each g_i is a group element in a vertex group of G and $\alpha_1 \dots \alpha_k$ is a loop in Γ . We then define

$$t' := g_0 t'_{\alpha_1} g_1 t'_{\alpha_2} \dots t'_{\alpha_k} g_k$$

which is a group element in G' . Note that $tv_i = t'v_i = v$ in the Bass–Serre tree T and hence $L'_v = L'_{t'v_i} = t'L'_{v_i}$ and $L_v = L_{tv_i} = tL_{v_i}$ by Remark 2.13. Since $L_v = tL_{v_i}$, it follows that each

element in L_v can be written as tx for some $x \in L_{v_i}$. This yields a well-defined quasi-isometry

$$\zeta_v : L_v \rightarrow L'_v, \quad tx \mapsto t' \zeta_{v_i}(x).$$

By finiteness of the G -orbits of vertices, the map ζ_v can be chosen uniformly quasi-isometric.

Definition 3.1. For each vertex v in $V(T)$, let $f_v : X_v \rightarrow \mathcal{H}_v \times L_v$ and $f'_v : X'_v \rightarrow \mathcal{H}'_v \times L'_v$ be the maps given by Lemma 2.16. Let e be an edge in $E(T)$ with $e_- = v$. We define the *vertex map*

$$\phi_v : X_v \rightarrow X'_v \quad \text{by} \quad \phi_v = (f'_v)^{-1} \circ (\text{id} \times \zeta_v) \circ f_v.$$

and the *edge map*

$$\phi_e : X_e \rightarrow X'_e \quad \text{by} \quad \phi_e = \left((f'_v)^{-1} \right) \Big|_{X'_e} \circ (\text{id} \times \zeta_v) \circ (f_v|_{X_e})$$

3.2. Proof of Theorem 1.3. In this section, we are going to prove Theorem 1.3. Given two admissible groups $\mathcal{G} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ and $\mathcal{G}' = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\sigma_{\hat{e}}\})$. Let $X := X(T, \{X_v\}, \{X_e\}, \{\alpha_e\})$ and $X' := X'(T', \{X'_v\}, \{X'_e\}, \{\alpha'_e\})$ be the tree of spaces associated to $\mathcal{G}, \mathcal{G}'$ given by Section 2.3. Here $\alpha_e : X_e \rightarrow X_{\bar{e}}$ and $\alpha'_e : X'_e \rightarrow X'_{\bar{e}}$.

Let $\{\phi_v\}$ and $\{\phi_e\}$ be the collection of vertex maps and edge maps given by Definition 3.1. We fix uniform constants $K \geq 1, A \geq 0$ so that each ϕ_v and ϕ_e is a (K, A) -quasi-isometry. The isomorphism $\xi : T \rightarrow T'$ here we are using is the identity $T \rightarrow T$.

For each vertex v in $V(T)$, let e be an edge such that $e_- = v$. By the construction of ϕ_v and ϕ_e in Section 3.1, the following diagram is commuted up to uniformly bounded error

$$\begin{array}{ccc} X_e & \xrightarrow{\phi_e} & X'_e \\ \downarrow i_e & & \downarrow i'_e \\ X_v & \xrightarrow{\phi_v} & X'_v \end{array}$$

Here i_e, i'_e are inclusion maps from edge spaces to vertex spaces. Hence we establish commutativity (up to uniformly bounded error) of the first sub-diagram in Theorem 2.6.

For the rest of the proof, we are going to verify that our maps satisfy the commutativity (up to uniformly bounded error) of the second sub-diagram in Theorem 2.6. In other words, if $w := e_+$ then we verify that the following diagram is commuted up to uniformly bounded error.

$$\begin{array}{ccc} X_e & \xrightarrow{\phi_e} & X'_e \\ \downarrow \alpha_e & & \downarrow \alpha'_e \\ X_v & \xrightarrow{\phi_w} & X'_w \end{array}$$

Claim 1: There exists a uniform constant $C > 0$ such that

- (1) For each oriented edge $e = [v, w]$, X_e is quasi-isometric to $L_w \times L_v$ via the following (C, C) -quasi-isometry $\rho_e : X_e \rightarrow L_w \times L_v$ defined by

$$x \mapsto (\psi_{\bar{e}} \circ \pi_v(x), \psi_e \circ \pi_w \circ \alpha_e(x)).$$

- (2) The following diagram is commuted up to an error C .

$$\begin{array}{ccc} X_e & \xrightarrow{\alpha_e} & X_{\bar{e}} \\ \downarrow \rho_e & & \downarrow \rho_{\bar{e}} \\ L_w \times L_v & \xrightarrow{\text{flip}(v, w)} & L_v \times L_w \end{array}$$

It is clear that (2) follows from (1). For (1), condition (4) of Definition 2.7 gives us

$$X_e \simeq_{q.i.} G_e \simeq_{q.i.} \langle Z_v, \alpha_{\bar{e}}(Z_w) \rangle = \alpha_{\bar{e}}(Z_w) \times Z_v.$$

Also, we can rewrite ρ_e as

$$x \mapsto (\psi_{\bar{e}} \circ \pi_v(x), \psi_e \circ \pi_w \circ \alpha_e(x)) = (p_w \circ \alpha_{\bar{e}}(x), p_v(x)),$$

where $p_v : X_v \rightarrow L_v, p_w : X_w \rightarrow L_w$. Since $Z_v \rightarrow L_v$ is a Z_v -equivariant quasi-isometry, the following map is a quasi-isometry

$$X_e \simeq_{q.i.} \alpha_{\bar{e}}(Z_w) \times Z_v \xrightarrow{p_w \times p_v} L_w \times L_v.$$

Claim 2: The following diagram commutes up to a uniform error.

$$\begin{array}{ccc} X_e & \xrightarrow{\phi_e} & X'_e \\ \downarrow i_{\bar{e}} \circ \alpha_e & & \downarrow i'_{\bar{e}} \circ \alpha'_e \\ X_w & \xrightarrow{\phi_w} & X'_w \end{array}$$

For our notations purpose, we write $f \approx g$ to mean two maps f and g are uniform close.

According to the diagram above Claim 1, we have

$$i'_{\bar{e}} \circ \phi_{\bar{e}} \circ \alpha_e \approx \phi_w \circ i_{\bar{e}} \circ \alpha_e$$

and hence to show that the above diagram is commuted up to a uniform error, it suffices to verify that

$$(1) \quad \phi_{\bar{e}} \circ \alpha_e \approx \alpha'_e \circ \phi_e$$

To see this, we consider the following diagram:

$$\begin{array}{ccccccc} X_v & \xleftarrow{i_e} & X_e & \xrightarrow{\alpha_e} & X_{\bar{e}} & \xrightarrow{i_{\bar{e}}} & X_w \\ \downarrow f_v & & \downarrow \rho_e & & \downarrow \rho_{\bar{e}} & & \downarrow f_w \\ Y_v \times L_v & \xleftarrow{\psi_{\bar{e}}^{-1} \times id} & L_w \times L_v & \xrightarrow{\text{flip}(v,w)} & L_v \times L_w & \xrightarrow{\psi_{\bar{e}}^{-1} \times id} & Y_w \times L_w \\ \downarrow id \times \zeta_v & & \downarrow \zeta_w \times \zeta_v & & \downarrow \zeta_v \times \zeta_w & & \downarrow id \times \zeta_w \\ Y_v \times L'_v & \xleftarrow{} & L'_w \times L'_v & \xrightarrow{\text{flip}'(v,w)} & L'_v \times L'_w & \longrightarrow & Y_w \times L'_w \\ \downarrow (f'_v)^{-1} & & \downarrow (\rho'_e)^{-1} & & \downarrow (\rho'_{\bar{e}})^{-1} & & \downarrow (f'_w)^{-1} \\ X'_v & \xleftarrow{i'_e} & X'_e & \xrightarrow{\alpha'_e} & X'_{\bar{e}} & \xrightarrow{i'_{\bar{e}}} & X'_w \end{array}$$

We note that:

- (1) the compositions of maps in the first and the fourth columns are ϕ_v and ϕ_w respectively;
- (2) the compositions of maps in the second and the third columns are uniformly close to ϕ_e and $\phi_{\bar{e}}$ respectively;
- (3) construction of maps ϕ_e and ϕ_v , together with Claim 1, shows that the sub-diagram in the diagram above either commutes or commutes up to a uniform error.

Therefore it is routine to chase around the above diagram to check that $\phi_{\bar{e}} \circ \alpha_e \approx \alpha'_e \circ \phi_e$, establishing

(1). Claim 2 is confirmed.

In conclusion, the collections $\{\phi_v\}_{v \in V(T)}$ and $\{\phi_e\}_{e \in E(T)}$ satisfy the hypotheses of Theorem 2.6. Therefore there is a quasi-isometry

$$\phi : X \rightarrow X'$$

such that $\phi|_{X_v} = \phi_v$ for every $v \in V(T)$.

4. ADMISSIBLE GROUPS ARE CAT(0) ON THE LARGE SCALE

In this section, we use Theorem 1.3 to prove Theorem 1.4.

4.1. Flip CKA action. We refer the reader to [CK02], [NY23] for the material recalled here.

Definition 4.1. We say that the action $G \curvearrowright X$ is *Croke-Kleiner admissible* (CKA) if G is an admissible group, and X is a Hadamard space (i.e., a complete proper CAT(0) space), and the action is geometric (i.e., properly and cocompactly by isometries). The space X is called the *admissible space* for the CKA action $G \curvearrowright X$.

Let $G \curvearrowright X$ be a Croke-Kleiner admissible action, where G is the fundamental group of an admissible graph of groups $\mathcal{G}$ and let $G \curvearrowright T$ be the action of G on the associated Bass-Serre tree of $\mathcal{G}$ (we refer the reader to Section 2.5 in [CK02] for a brief discussion). Let $T^0 = \text{Vertex}(T)$ and $T^1 = \text{Edge}(T)$ be the vertex and edge sets of T . For each $\sigma \in T^0 \cup T^1$, we let $G_\sigma \leq G$ be the stabilizer of σ . For each vertex $v \in T^0$, let $Y_v := \text{Minset}(Z(G_v)) := \cap_{g \in Z(G_v)} \text{Minset}(g)$ and for every edge $e \in E$ we let $Y_e := \text{Minset}(Z(G_e)) := \cap_{g \in Z(G_e)} \text{Minset}(g)$. We note that the assignments $v \mapsto Y_v$ and $e \mapsto Y_e$ are G -equivariant with respect to the natural G actions.

We recall some facts from [CK02, Section 3.2] and [NY23, Section 2].

- (1) G_v acts co-compactly on $Y_v = \bar{Y}_v \times \mathbb{R}$ and $Z(G_v)$ acts by translation on the $\mathbb{R}$ -factor and trivially on $\bar{Y}_v$ where $\bar{Y}_v$ is a Hadamard space.
- (2) $G_e = \mathbb{Z}^2$ acts co-compactly on $Y_e = \bar{Y}_e \times \mathbb{R}^2 \subset Y_v$ where $\bar{Y}_e$ is a compact Hadamard space.
- (3) if $\langle t_1 \rangle = Z(G_{v_1})$, $\langle t_2 \rangle = Z(G_{v_2})$ then $\langle t_1, t_2 \rangle$ is a finite index subgroup of G_e .

We first choose, in a G -equivariant way, a plane $F_e \subset Y_e$ for each $e \in T^1$.

Definition 4.2 (Flip CKA action). If for each edge $e := [v, w] \in T^1$, the boundary line $\ell = \bar{Y}_v \cap F_e$ is parallel to the $\mathbb{R}$ -line in $Y_w = \bar{Y}_w \times \mathbb{R}$, then the CKA action is called *flip*.

4.2. Proof of Theorem 1.4. In this section, we are going to prove Theorem 1.4. We first review some results that will be used.

In [Wis00], Wise introduces the concept of an *omnipotent group* which has been widely used in subgroup separability.

Definition 4.3. A set of group elements $h_1, \dots, h_r$ in a group H is called *independent* if whenever h_i and h_j have conjugate powers then $i = j$. A group H is *omnipotent* if whenever $\{h_1, \dots, h_r\}$ ($r \geq 1$) is an independent set of group elements, then there is a positive integer $p > 1$ such that for every choice of positive integers $\{n_1, \dots, n_r\}$, there is a finite quotient $\varphi: H \rightarrow \bar{H}$ such that $\varphi(h_i)$ has order $n_i p$ in $\bar{H}$ for each i .

It is worth mentioning that free groups [Wis00], surface groups [Baj07], Fuchsian groups [Wil10] and virtually special hyperbolic groups [Wis00] all belong to the omnipotent group category. However, it is a longstanding open question whether every hyperbolic group is residually finite. Wise suggested that if every hyperbolic group is residually finite, then any hyperbolic group would be considered an omnipotent group [Wis00, Remark 3.4].

By [BH99, Theorem II.6.12], each vertex group $G_{\hat{v}}$ of the admissible group G contains a subgroup $K_{\hat{v}}$ intersecting trivially with $Z_{\hat{v}}$ so that the direct product $K_{\hat{v}} \times Z_{\hat{v}}$ is a finite subgroup of $G_{\hat{v}}$. The image of $K_{\hat{v}}$ in the quotient $Q_{\hat{v}} = G_{\hat{v}}/Z_{\hat{v}}$ is of finite index of $Q_{\hat{v}}$. Since $Q_{\hat{v}}$ is omnipotent and then is residually finite, we can assume that $K_{\hat{v}}$ is torsion-free.

A collection of finite index subgroups $\{G'_{\hat{e}}, G'_{\hat{v}} \mid \hat{v} \in V(\Gamma), \hat{e} \in E(\Gamma)\}$ of vertex and edge groups of $G = \pi_1(\mathcal{G})$ is called *compatible* if $G'_{\hat{e}} = \tau_e(G'_{\hat{e}})$ and whenever $\hat{v} = \hat{e}_-$ we have $G'_{\hat{e}} = G'_{\hat{v}} \cap G_{\hat{e}}$. When studying the virtual properties of a graph of groups G , it is frequently necessary to create a finite

index subgroup G' from a set of finite index subgroups of vertex groups. This can be accomplished using the following theorem.

Theorem 4.4. [DK18, Theorem 7.50] *Let G be the fundamental group of a graph of groups $\mathcal{G} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$. For every compatible collection $\{G'_{\hat{e}}, G'_{\hat{v}} \mid \hat{v} \in V(\Gamma), \hat{e} \in E(\Gamma)\}$ of $\mathcal{G}$, there exists a finite index subgroup $G' < G$ such that $G' \cap G_{\hat{v}} = G'_{\hat{v}}$ and $G' \cap G_{\hat{e}} = G'_{\hat{e}}$ for every vertex $\hat{v}$ and edge $\hat{e}$.*

Lemma 4.5. [HNY23, Lemma 4.8] *Let $\{K_{\hat{v}} \leq K_{\hat{v}} : v \in V(\Gamma)\}$ be a collection of finite index subgroups. Then there exist finite index subgroups $\tilde{K}_{\hat{v}}$ of $K_{\hat{v}}$, $G'_{\hat{e}}$ of $G_{\hat{e}}$ and $Z'_{\hat{v}}$ of $Z_{\hat{v}}$ so that the collection of finite index subgroups $\{G'_{\hat{e}}, G'_{\hat{v}} = \tilde{K}_{\hat{v}} \times Z'_{\hat{v}} : v \in V(\Gamma), e \in E(\Gamma)\}$ is compatible.*

We are now ready for the proof of Theorem 1.4. We recall the statement of Theorem 1.4 for the convenience of the reader.

Theorem 1.4. *Let G be an admissible group such that each vertex group is a central extension of an omnipotent hyperbolic $CAT(0)$ group by $\mathbb{Z}$. Then G is quasi-isometric to a group admitting a flip CKA action.*

Proof. According to [NY23, Lemma 4.6], there is a subgroup of G that has a finite index of at most 2 and is also an admissible group, with a bipartite underlying graph. For simplicity, we still refer to this subgroup as G . Using Lemma 4.5, we obtain an admissible group $\mathcal{G}' = (\Gamma', \{G'_{\hat{u}} = K_{\hat{u}} \times \mathbb{Z}_{\hat{u}}, \{G'_{\hat{e}}, \{\tau_{\hat{e}}\}\})$ where

- (1) $K_{\hat{u}}$ is a torsion-free, omnipotent $CAT(0)$, nonelementary hyperbolic group.
- (2) $G' := \pi_1(\mathcal{G}')$ is a finite index subgroup of G .

For each vertex $\hat{u}_i \in V(\Gamma')$, let $Y_{\hat{u}_i}$ be a $CAT(0)$ hyperbolic space such that $K_{\hat{u}_i} \curvearrowright Y_{\hat{u}_i}$ geometrically. Fix a generator $t_{\hat{u}_i}$ of the factor $\mathbb{Z}_{\hat{u}_i}$ of $K_{\hat{u}_i} \times \mathbb{Z}_{\hat{u}_i}$. Then $G'_{\hat{u}_i} = K_{\hat{u}_i} \times \langle t_i \rangle$ acts geometrically on the $CAT(0)$ space $X_{\hat{u}_i} := Y_{\hat{u}_i} \times \mathbb{R}$. Let $\hat{e}$ be an oriented edge in Γ' such that $\hat{e}_- = \hat{u}_i$. The image $\pi'_{\hat{u}_i}(G'_{\hat{e}}) \leq K_{\hat{u}_i}$ under the projection $\pi'_{\hat{u}_i} : G'_{\hat{u}_i} \rightarrow K_{\hat{u}_i}$ is an infinite cyclic subgroup generated by an element $k_{\hat{e}} \in K_{\hat{u}_i}$. The hyperbolic element $k_{\hat{e}}$ gives rise to a totally geodesic torus $T_{\hat{e}}$ in the quotient space $X_{\hat{u}_i}/G'_{\hat{u}_i}$ with basis denoted by $([k_{\hat{e}}], [t_{\hat{u}_i}])$. We re-scale $Y_{\hat{u}_i}$ so that the translation length of $k_{\hat{e}}$ is equal to that of $t_{\hat{u}_i}$ for each i . Let

$$f_{\hat{e}} : T_{\hat{e}} \rightarrow T_{\hat{e}}$$

be a flip isometry respecting these lengths, that is, an orientation-reversing isometry mapping $[k_{\hat{e}}]$ to $[t_{\hat{e}-}]$ and $[t_{\hat{e}+}]$ to $[k_{\hat{e}}]$.

Let M be the space obtained from taking the disjoint union of compact spaces $\bigsqcup_{\hat{u}_i \in V(\Gamma')} X_{\hat{u}_i}/G'_{\hat{u}_i}$ and glue these spaces accordingly via isometry $f_{\hat{e}} : T_{\hat{e}} \rightarrow T_{\hat{e}}$ with $\hat{e}$ varies oriented edges on the underlying graph Γ' .

The fundamental group $\pi_1(M)$ has a graph of groups structure as follows:

- for each vertex $\hat{u}_i$, the associated vertex group is $\pi_1(X_{\hat{u}_i}/G'_{\hat{u}_i})$;
- for each oriented edge $\hat{e}$, the associated edge group is $\pi_1(T_{\hat{e}})$. Edge monomorphisms are $(f_{\hat{e}})_* : \pi_1(T_{\hat{e}}) \rightarrow \pi_1(T_{\hat{e}})$ induced by $f_{\hat{e}} : T_{\hat{e}} \rightarrow T_{\hat{e}}$.

There is a metric on M which makes M into a locally $CAT(0)$ space (see e.g. [BH99, Proposition II.11.6]).

Let $\tilde{M} \rightarrow M$ be the universal cover of M . By the Cartan-Hadamard Theorem, the universal cover $\tilde{M}$ with the induced length metric from M is a $CAT(0)$ space, and hence $\pi_1(M)$ is a $CAT(0)$ admissible groups as $\pi_1(M)$ acts geometrically on $\tilde{M}$.

As two admissible groups G' and $\pi_1(M)$ have the same underlying graph, same vertex groups, and same edge groups. The only difference is gluing edge maps. We thus can apply Theorem 1.3

to conclude that G' and $\pi_1(M)$ are quasi-isometric, and hence G is quasi-isometric to $\pi_1(M)$ since G' is a finite index subgroup of G . $\square$

Below, we give the proof of Corollary 1.5. We need several lemmas.

Lemma 4.6. [BD14, Corollary 4.17] *If a finitely generated group G is strongly thick of order at most n , then the divergence of G is bounded above by a polynomial of degree $n + 1$.*

Lemma 4.7. [BD14, Theorem 6.4] *Let γ be a Morse quasi-geodesic in a $CAT(0)$ metric space X . Then the divergence of X is at least quadratic.*

Suppose that G contains a vertex group of type $\mathcal{H}$. By the normal form theorem, for each connected subgraph Γ' of Γ , there is a subgroup $G_{\Gamma'} \leq G$ which is the fundamental group of the graph of groups with underlying graph Γ' , and with vertex, edge groups, and edge monomorphisms coming from $\mathcal{G}$. Let Λ be the full subgraph of Γ with vertex set $\{v \in V\Gamma : G_v \text{ is type } \mathcal{S}\}$. For each component Γ' of Λ , we say that $G_{\Gamma'}$ is

- (1) a *maximal admissible component* if Γ' contains an edge;
- (2) an *isolated type $\mathcal{S}$ vertex group* if Γ' consists of a single vertex of type $\mathcal{S}$.

Recall that if G_v is a vertex group of type $\mathcal{H}$, then it is a relatively hyperbolic group to $\mathbb{P}_v$.

We remark that every graph of groups is obtained by iterating amalgamated products and HNN extensions. By applying the Combination Theorem of relatively hyperbolic groups [Dah03, Theorem 0.1] to our setting $\mathcal{G}'$, specifically (2) and (3) of [Dah03, Theorem 0.1] for amalgamated products and (4) for HNN extensions, we obtain the following:

Lemma 4.8. [Ngu25, Lemma 4.1] *Let $G_1, \dots, G_k$ be the maximal admissible components and isolated vertex groups of type $\mathcal{S}$ of an extended admissible group G . Let $G_{e_1}, \dots, G_{e_m}$ be the edge groups so that both its associated vertex groups $G_{(e_i)_{\pm}}$ are of type $\mathcal{H}$, and let $T_1, \dots, T_\ell$ be groups in $\cup \mathbb{P}_v$ which are not edge groups of G . Then G is hyperbolic relative to*

$$\mathbb{P} = \{G_i\}_{i=1}^k \cup \{G_{e_s}\}_{s=1}^m \cup \{T_i\}_{i=1}^\ell$$

Corollary 1.5. *Let G be an extended admissible group such that for each vertex group G_v of type $\mathcal{S}$, its non-elementary hyperbolic factor Q_v is omnipotent and is a $CAT(0)$ group. Then the divergence of G is quadratic if and only if G contains no vertex groups of type $\mathcal{H}$ and it is exponential otherwise.*

In particular, assume that G' is another extended admissible group satisfying the same conditions as G . If G contains no vertex groups of type $\mathcal{H}$ and G' contains at least one vertex group of type $\mathcal{H}$, then G and G' are not quasi-isometric.

Proof. We consider the following two cases.

Case 1: G contains no vertex group of type $\mathcal{H}$. In this case G is an admissible group. We first show that the upper bound of the divergence is quadratic. By [MN24, Corollary 3.11], the inclusion of a vertex group $G_v \rightarrow G$ is a quasi-isometric embedding, and hence for any two points $x, y \in G_v$, a geodesic γ in G_v connecting x to y will be a uniform quasi-geodesic in G . This shows that the graph G_v satisfies the quasi-convexity property as defined in [BD14, §4.1]. Since every asymptotic cone of a vertex group of G is without cut-points, it follows that vertex groups of G are strongly algebraically thick of order zero in the sense of [BD14]. We have that G is strongly thick of order at most 1 since a graph of groups with infinite edge groups and whose vertex groups is thick of order n , is thick of order at most $n + 1$, by [BD14, Proposition 4.4 & Definition 4.14]. Using Lemma 4.6, we have that the divergence of G is at most quadratic.

Now we consider the lower bound of the divergence. According to Theorem 1.4, there exists a $CAT(0)$ admissible group G' so that G and G' are quasi-isometric. Pick any infinite order group element $g \in G'$ which is not conjugate into any vertex group of G' . Then by [NY23, Corollary 6.16], g is a Morse element in G' . According to Lemma 4.7, the divergence of G' is at least quadratic, and

hence the divergence of G must be at least quadratic since divergence is a quasi-isometric invariant [Ger94]. Therefore the divergence of G is quadratic.

Case 2: G contains at least one vertex group of type $\mathcal{H}$. In this case G has the natural relatively hyperbolic structure described by Lemma 4.8. According to [Sis12, Theorem 1.3], the divergence of a relatively hyperbolic group is exponential, and hence the divergence of G is exponential. $\square$

Corollary 1.6. *Let G be an extended admissible group such that for each vertex group G_v of type S , its non-elementary hyperbolic factor Q_v is omnipotent and is a $CAT(0)$ group. Then G has well-defined quasi-redirecting boundary.*

Proof. We consider the following two cases.

Case 1: G contains no vertex group of type $\mathcal{H}$. In this case, we apply Theorem 1.4 to obtain a $CAT(0)$ admissible group G' so that G and G' are quasi-isometric. In [NQ25, Section 5], the authors prove that for the $CAT(0)$ admissible group G' , its quasi-redirecting boundary is well-defined. Since quasi-redirecting boundary is a quasi-isometric invariant, it follows that G has well-defined quasi-redirecting boundary.

Case 2: G contains at least one vertex group of type $\mathcal{H}$. Let $\mathbb{P} = \{G_i\}_{i=1}^k \cup \{G_{e_s}\}_{s=1}^m \cup \{T_i\}_{i=1}^\ell$ be the peripheral subgroups of G in the relatively hyperbolic structure of G given by Lemma 4.8. Each T_i and G_{e_s} are quasi-isometric to $\mathbb{R}^2$ so they do have well-defined quasi-redirecting boundary. The well-defined quasi-redirecting boundary of each G_i is confirmed by Case 1. Thus we have shown that each peripheral subgroup in this relatively hyperbolic structure has well-defined quasi-redirecting boundary, and hence it follows from [NQ25, Theorem D] that G has well-defined quasi-redirecting boundary. $\square$

Corollary 1.7. *Let Φ be a linearly growing automorphism of the finite rank free group F and let $G = F \rtimes_{\Phi} \langle t \rangle$ be its mapping torus. Suppose that G is unbranched in the sense of [BGGH25]. Then G has well-defined quasi-redirecting boundary.*

Proof. It is shown in the proof of [BGGH25, Lemma 6.8] that there is a finite index subgroup Γ of G such that Γ is an admissible group Γ where each vertex group Γ_v of Γ is a direct product of a free group F_v with $\mathbb{Z}$. Note that Γ_v is a $CAT(0)$ group and F_v is omnipotent. It follows from Corollary 1.6 that Γ has well-defined quasi-redirecting boundary. Since quasi-redirecting boundary is well-behaved under quasi-isometries, it follows that G has well-defined quasi-redirecting boundary as Γ has finite index in G . $\square$

5. SUBGROUPS OF EXTENDED ADMISSIBLE GROUPS

In this section, we study various aspects of subgroups of extended admissible groups.

Proposition 5.1. *Let G be an extended admissible group. Then G has Rapid Decay property. As a consequence, the following groups can not be embedded in extended admissible groups.*

- Amenable groups with exponential growth.
- The Baumslag–Solitar group $BS(p, q) = \langle a, b \mid ba^p b^{-1} = a^q \rangle$ where $p \neq \pm q$, $p, q \in \mathbb{Z}$ are nonzero integers.
- Thompson’s group, $SL_n(\mathbb{Z})$ with $n \geq 3$, intermediate growth groups.

Proof. We consider the following two cases.

Case 1: G contains no vertex group of type $\mathcal{H}$. In this case, each vertex group is a central extension of a hyperbolic group, hence it is Rapid Decay ([Nos92]). Instead of recalling the precise definition of Rapid Decay, we refer the reader to [Cha17] for a clear survey on this property, since we only require some of its basic properties. Also, both vertex groups and edge groups are quasi-isometric embedded [MN24, Lemma 2.6, Corollary 3.11], hence these groups are undistorted in G

and they have loose polynomial distortion [CG24]. Applying [CG24, Proposition 1.3], G has Rapid Decay property.

Case 2: G contains a vertex group of type $\mathcal{H}$. In this case, G is hyperbolic relative to

$$\mathbb{P} = \{G_i\}_{i=1}^k \cup \{G_{e_s}\}_{s=1}^m \cup \{T_i\}_{i=1}^l,$$

which is shown in Lemma 4.8. Each vertex group G_n of type $\mathcal{H}$ is relative to $\mathbb{P}_n$, and each group in $\mathbb{P}_n$ is Rapid Decay [Jol90, Theorem 3.1.7], hence G_n is also Rapid Decay by [DS05, Theorem 1.1]. Since G is relatively hyperbolic to $\mathbb{P}$, by applying [DS05, Theorem 1.1] again, G has the Rapid Decay property.

We remark here that the Rapid Decay property is preserved by passing to subgroups. In [Cha17], the author lists several classes of groups which do not have the Rapid Decay property, including amenable groups with exponential growth, Baumslag–Solitar groups, Thompson’s group, $SL_n(\mathbb{Z})$ with $n \geq 3$, and groups of intermediate growth. $\square$

Definition 5.2. Let G be a group. A subgroup H of G is *separable* if and only if for all $g \in G \setminus H$, there exists a finite index subgroup $K < G$ such that $H < K < G$ and $g \notin K$. The group G is called *locally extended residually finite* (LERF) if any finitely generated subgroup of G is separable.

We consider the following Croke–Kleiner group:

$$L = \langle i, j, k, l \mid [i, j], [j, k], [k, l] \rangle.$$

The group L is the fundamental group of a torus complex (see Example 2.9). Note that it is also the fundamental group of a graph manifold and this is a right-angled Artin group on the line graph with four vertices and three edge. This group appears in [CK00] as an example of admissible groups used to show the existence of a group acting geometrically on distinct CAT(0) spaces whose visual boundaries are not homeomorphic (see [CK00]). This observation has served as motivation for several boundary constructions in recent years.

In [NW01], the authors prove the following.

Lemma 5.3. [NW01, Theorem 1.2] *The Croke–Kleiner group L is not LERF.*

Proposition 5.4. *Suppose that an extended admissible group G contains at least one maximal admissible component then there is an embedding $L \rightarrow G$. In particular, G is not LERF.*

Proof. It suffices to consider G as an admissible group, since any group containing a non-LERF subgroup is itself not LERF.

According to [NY23, Lemma 4.6], there is a subgroup $\dot{G}$ of G that has a finite index of at most 2 and is also an admissible group, with a bipartite underlying graph. We denote the graph of groups structure of $\dot{G}$ by $\mathcal{K} = (\Gamma, \{G_{\hat{v}}\}, \{G_{\hat{e}}\}, \{\tau_{\hat{e}}\})$ where $\tau_{\hat{e}}$ is an isomorphism $G_{\hat{e}} \rightarrow G_{\hat{e}^-}$.

Pick an edge $\hat{e}$ in the underlying graph of the admissible group $\dot{G}$ with two distinct vertices $\hat{v} = \hat{e}_-$ and $\hat{w} = \hat{e}_+$. Choose a generator $\xi_{\hat{v}}$ of $Z_{\hat{v}} := Z(\dot{G}_{\hat{v}}) \cong \mathbb{Z}$, and choose a generator $\xi_{\hat{w}}$ of $Z_{\hat{w}} := Z(\dot{G}_{\hat{w}}) \cong \mathbb{Z}$. We recall that the subgroup generated by $Z_{\hat{v}}$ and $\tau_{\hat{e}}(Z_{\hat{w}})$ has finite index in $G_{\hat{e}} \cong \mathbb{Z}^2$ (see (4) in Definition 2.7). Since $\xi_{\hat{v}}$ is not contained in $Z_{\hat{w}}$, there exists an element $h_{\hat{w}}$ in $G_{\hat{w}}$ so that $\xi_{\hat{v}}$ does not commute with $h_{\hat{w}}$. Similarly, there exists an element $h_{\hat{v}}$ in $G_{\hat{v}}$ so that $\xi_{\hat{w}}$ does not commute with $h_{\hat{v}}$. Since $\xi_{\hat{v}} \in Z_{\hat{v}}$ and $\xi_{\hat{w}} \in Z_{\hat{w}}$, we have that $[\xi_{\hat{v}}, h_{\hat{v}}] = 1$, $[\xi_{\hat{w}}, h_{\hat{w}}] = 1$ and $[\xi_{\hat{v}}, \xi_{\hat{w}}] = 1$. Let $\check{G} := \langle h_{\hat{v}}, \xi_{\hat{v}}, \xi_{\hat{w}}, h_{\hat{v}} \rangle$ be the subgroup of $\dot{G}$. We consider the map $\psi: L \rightarrow \check{G}$ given by

$$i \mapsto h_{\hat{v}}, j \mapsto \xi_{\hat{v}}, k \mapsto \xi_{\hat{w}}, l \mapsto h_{\hat{w}}$$

Since $\psi(r) = 1$ for every relator r in L , it follows that ψ is a homomorphism. Normal forms show that the homomorphism ψ is injective. Thus there is an embedding $L \rightarrow \check{G}$. Since L is not LERF (see Lemma 5.3), it follows that $\check{G}$ is not LERF. $\square$

Definition 5.5. Recall that a subgroup $H < G$ is *malnormal* if $H \cap gHg^{-1}$ is trivial for all $g \notin H$, and is *almost malnormal* if $H \cap gHg^{-1}$ is finite for all $g \notin H$. Let $H \leq G$. The *height* of H in G is the largest number $n > 0$ so that there are n distinct cosets $\{g_1H, g_2H, \dots, g_nH\}$ so that the intersection of conjugates $g_iHg_i^{-1}$ is infinite. Thus finite groups have height 0, infinite almost malnormal subgroups have height 1, and so on.

Definition 5.6. Let G be a finitely generated group and H a subgroup of G . We say H is *strongly quasiconvex* in G if for any $L > 1$, $C > 0$ there exists $M = M(L, C)$ such that every (L, C) -quasi-geodesic in G with endpoints in H is contained in the M -neighborhood of H .

In [Tra19], Tran shows that strongly quasiconvex subgroups in a finitely generated group have finite height. While the equivalence of strong quasiconvexity and finite height has been established for extended admissible groups in [Ngu24], the relationship with virtual malnormality in the context of extended admissible groups has not been explicitly treated.

In the setting of relatively hyperbolic groups, Hruska-Wise in [HW09] prove the following result mentioning that this result is new even in the hyperbolic case. Here we recall a subgroup H of G is *separable* if and only if for all $g \in G \setminus H$, there exists a finite-index subgroup $K \leq G$ such that $H \leq K \leq G$ and $g \notin K$.

Proposition 5.7. [HW09, Theorem 9.3] *Let H be a separable, relatively quasiconvex subgroup of the relatively hyperbolic group G . Then there is a finite index subgroup K of G containing H such that H is relatively malnormal in K .*

We generalize this result to a broader setting by showing that strongly quasiconvex and separable subgroups are virtually almost malnormal. This result applies to the setting of extended admissible groups and may be of independent interest.

Theorem 1.8. *Let H be a separable, strongly quasiconvex subgroup of a finitely generated group G . Then there is a finite index subgroup K of G containing H such that H is almost malnormal in K . Furthermore, suppose that G is virtually torsion-free then H is virtually malnormal.*

Suppose G is the fundamental group of a graph 3-manifold M . Then a finitely generated subgroup H of $\pi_1(M)$ is strongly quasiconvex if and only if H is virtually malnormal in G .

Proof. We first claim that there are only finitely many double cosets $Hg_1H, Hg_2H, \dots, Hg_nH$ such that $H \cap g_iHg_i^{-1}$ is infinite. Indeed, suppose $\{g_i | i \in I\}$ is a collection of cosets such that $H \cap g_iHg_i^{-1}$ is infinite for each i . We fix a finite generating set S of G . By the proof of Theorem 4.15 in [Tra19] there is a constant C such that $d_S(H, g_iH) < C$ for each i . Thus we can translate g_iH by an element of H to obtain a coset hg_iH intersecting the ball of radius C in the Cayley graph $\Gamma(G, S)$ centered at the identity. Since this ball is finite, it follows that the cosets g_iH lie in only finitely many double cosets Hg_iH .

Since H is separable, there exists a finite index subgroup K of G containing H and $g_i \notin K$ for each i . If $k \in K - H$ and $H \cap kHk^{-1}$ is infinite, then $kH = hg_iH$ for some g_i and some $h \in H$. Also H is a subgroup of K . Therefore, g_i is a group element in K , contradicting our choice of K . Consequently H is almost malnormal in K .

Suppose that G is virtually torsion-free. $G_1 < G$ be the torsion free finite index subgroup of G and let $G_2 = K \cap G_1$. We then have $H_2 = H \cap G_2$ malnormal in G_2 using [AGM16, Lemma 4.24].

Now we assume that G is the fundamental group of a graph 3-manifold M . If H is virtually malnormal then H is virtually finite height and hence H is strongly quasiconvex in $\pi_1(M)$ by [NTY21]. Now we assume that H is strongly quasiconvex in $\pi_1(M)$. By [NS20] H must be separable in $\pi_1(M)$ since otherwise the distortion of H in $\pi_1(M)$ is quadratic or exponential which contradicts to the fact H is strongly quasiconvex in $\pi_1(M)$. Therefore H is virtually malnormal in $\pi_1(M)$. $\square$

REFERENCES

- [AGM16] I. Agol, D. Groves, J. Manning. *An alternate proof of Wise’s Malnormal Special Quotient Theorem*. Forum of Mathematics, Pi 4, e1 (2016).
- [AFW15] M. Aschenbrenner, S. Friedl, H. Wilton. *3-manifold groups*. EMS Series of Lectures in Mathematics. European Mathematical Society (EMS), Zürich, 2015.
- [ANR24] C. Abbott, H. Nguyen, A. Rasmussen. *Largest hyperbolic actions of 3-manifold groups*. Bull. London Math. Soc., 56 (2024):3090–3013.
- [Baj07] J. Bajpai. *Omnipotence of surface groups*. Master’s thesis, McGill University, 2007.
- [BD14] J. Behrstock, C. Drutu. *Divergence, thick groups, and short conjugators*. Illinois J. Math., 58 (2014), no. 4, 939–980.
- [BN08] J. Behrstock, W. Neumann. *Quasi-isometric classification of graph manifold groups*. Duke Math. J., 141(2):217–240, 2008.
- [BGH25] E. Bongiovanni, P. Ghosh, F. Gultepe, M. Hagen. *Characterizing hierarchically hyperbolic free by cyclic groups*. <https://arxiv.org/abs/2508.15738>
- [BH99] M. Bridson, A. Haefliger. *Metric spaces of non-positive curvature*. Springer-Verlag, 1999.
- [CG24] I. Chatterji, F. Gautero. *Distortion in graphs of groups and rapid decay classification of 3-manifold groups*. J. Noncommut. Geom. (2024), published online first.
- [Cha17] I. Chatterji. *Introduction to the Rapid Decay property*. Contemporary Mathematics 2017, Volume 691, 55–72.
- [CK00] C. Croke, B. Kleiner. *Spaces with nonpositive curvature and their ideal boundaries*. Topology, 39(3):549–556, 2000.
- [CK02] C. Croke, B. Kleiner. *The geodesic flow of a nonpositively curved graph manifold*. Geom. Funct. Anal., 12(3):479–545, 2002.
- [CM17] C. Cashen, A. Martin. *Quasi-isometries between groups with two-ended splittings*. Math. Proc. Camb. Phil. Soc., 162(2):249–291, 2017.
- [CS15] R. Charney, H. Sultan. *Contracting boundaries of CAT(0) spaces*. J. Topol., 8(1):93–117, 2015.
- [Dah03] F. Dahmani. *Combination of convergence groups*. Geom. Topol., 7:933–963, 2003.
- [DK18] C. Drutu, M. Kapovich. *Geometric group theory*. Amer. Math. Soc. Colloq. Publ. 63, AMS, 2018.
- [DS05] C. Drutu, M. Sapir. *Relatively hyperbolic groups with Rapid Decay property*. Int. Math. Res. Not., no. 19, 1181–1194, 2005.
- [Ger94] S. Gersten. *Quadratic divergence of geodesics in CAT(0) spaces*. Geom. Funct. Anal., 4(1):37–51, 1994.
- [Ger94a] S. Gersten. *Divergence in 3-manifold groups*. Geom. Funct. Anal., 4(6):633–647, 1994.
- [HNY23] S. Han, H. Nguyen, W. Yang. *Property (QT) for 3-manifold groups*. Alg. Geom. Topol., 25 (2025), 107–159.
- [HRSS22] M. Hagen, J. Russell, A. Sisto, D. Spriano. *Equivariant hierarchically hyperbolic structures for 3-manifold groups via quasimorphisms*. Ann. Inst. Fourier. 75:2, 769–828, 2025.
- [HW09] C. Hruska, D. Wise. *Packing subgroups in relatively hyperbolic groups*. Geom. Topol., 13 (2009) 1945–1988.
- [Jol90] P. Jolissaint. *Rapidly decreasing functions in reduced C^* -algebra of groups*. Trans. Amer. Math. Soc., 317:167–196, 1990.
- [KL97] M. Kapovich, B. Leeb. *Quasi-isometries preserve the geometric decomposition of Haken manifolds*. Invent. Math., 128(2):393–416, 1997.
- [KL98] M. Kapovich, B. Leeb. *3-manifold groups and nonpositive curvature*. Geom. Funct. Anal., 8(5):841–852, 1998.
- [MN24] A. Margolis, H. Nguyen. *Quasi-isometric rigidity of extended admissible groups*. <https://arxiv.org/abs/2401.03635>.
- [Ngu24] H. Nguyen. *Finite height subgroups of extended admissible groups*. New York Journal of Mathematics., 30 (2024) 1220–1234.
- [Ngu25] H. Nguyen. *Separability properties of extended admissible groups*. To appear in Forum Math. <https://www.degruyterbrill.com/document/doi/10.1515/forum-2024-0194/html>
- [Nos92] G. Noskov. *The algebra of Rapid Decay functions on groups and cocycle of polynomial growth*. Sib. Math. J., 33:634–640, 1992.
- [NQ24] H. Nguyen, Y. Qing. *Sublinearly Morse boundary of CAT(0) admissible groups*. J. Group Theory, 27(4):857–897, 2024.
- [NQ25] H. Nguyen, Y. Qing. *Quasi-redirecting boundaries of nonpositively curved groups*. <https://arxiv.org/abs/2503.22994>.
- [NS20] H. Nguyen, H. Sun. *Subgroup distortion of 3-manifold groups*. Trans. Amer. Math. Soc, 373 (2020), no. 9, 6683–6711.
- [NTY21] H. Nguyen, H. Tran, W. Yang. *Quasiconvexity in 3-manifold groups*. Math. Ann., 381(1-2):405–437, 2021.

- [NY23] H. Nguyen, W. Yang. *Croke–Kleiner admissible groups: Property (QT) and quasiconvexity*. Mich. Math. J., 73(5):971–1019, 2023.
- [NW01] G. Niblo, D. Wise. *Subgroup separability, knot groups and graph manifolds*. Proceedings of the American Mathematical Society 129 (3), 685–693.
- [Osin06] D. Osin. *Relatively hyperbolic groups: intrinsic geometry, algebraic properties, and algorithmic problems*. Mem. Amer. Math. Soc., 179(843):vi+100, 2006.
- [QR24] Y. Qing, K. Rafi. *The quasi-redirecting boundary*. <https://arxiv.org/abs/2406.16794>
- [Sis12] A. Sisto. *On metric relative hyperbolicity*. <https://arxiv.org/abs/1210.8081>
- [Sun19] H. Sun. *Non-LERFness of arithmetic hyperbolic manifold groups and mixed 3-manifold groups*. Duke Math. J. 168 (2019), no.4, 655–696.
- [SW79] P. Scott, T. Wall. *Topological methods in group theory*. London Math. Soc. Lect. Note Ser. 36, 137–203, 1979.
- [Tra19] H. Tran. *On strongly quasiconvex subgroups*. Geom. Topol., 23(3):1173–1235, 2019.
- [Wil10] H. Wilton. *Virtual retractions, conjugacy separability and omnipotence*. J. Algebra, 323(2):323–335, 2010.
- [Wis00] D. Wise. *Subgroup separability of graphs of free groups with cyclic edge groups*. Q. J. Math., 51(1):107–129, 2000.

FACULTY OF MATHEMATICS AND COMPUTER SCIENCE, UNIVERSITY OF SCIENCE, HO CHI MINH CITY, VIETNAM

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY, VIETNAM

Email address: daotrongtoan.dtt4@gmail.com

FPT UNIVERSITY, DA NANG, VIETNAM

Email address: nthoang.math@gmail.com